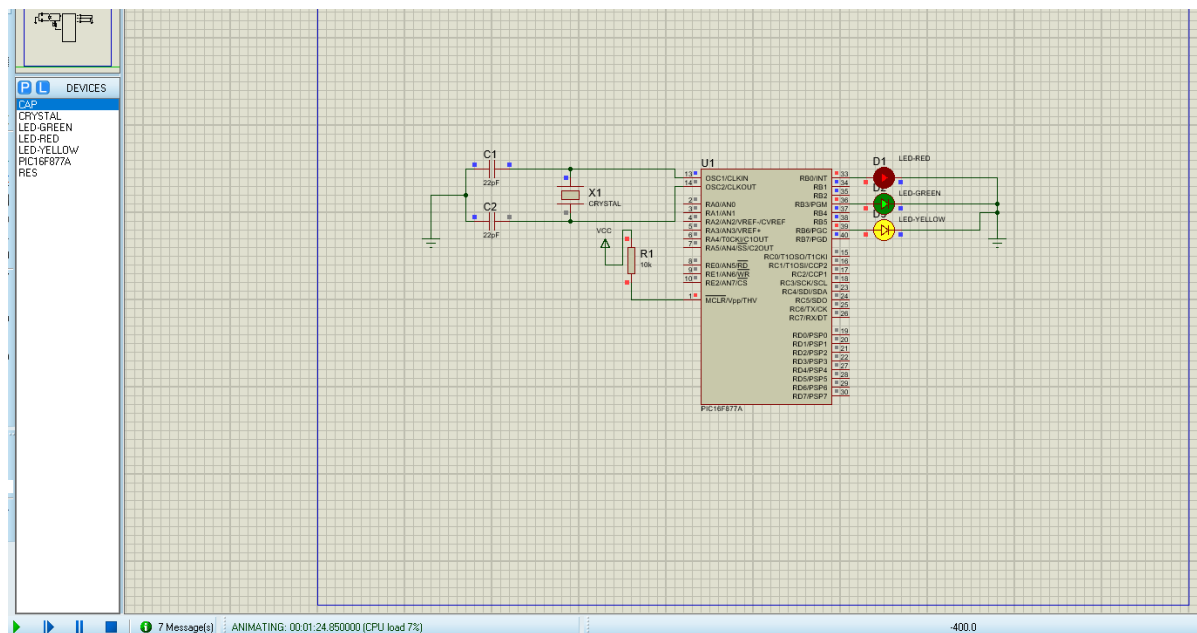


Source Code : LED blinking

```
void main() {  
    TRISB=0x00 ; // set portb as output  
    portb=0x00 ; //initialize portb as off  
    while (1){  
        portb.f0=1;  
        delay_ms(1000);  
        portb.f0=0;  
        delay_ms(1000);  
    }  
}
```

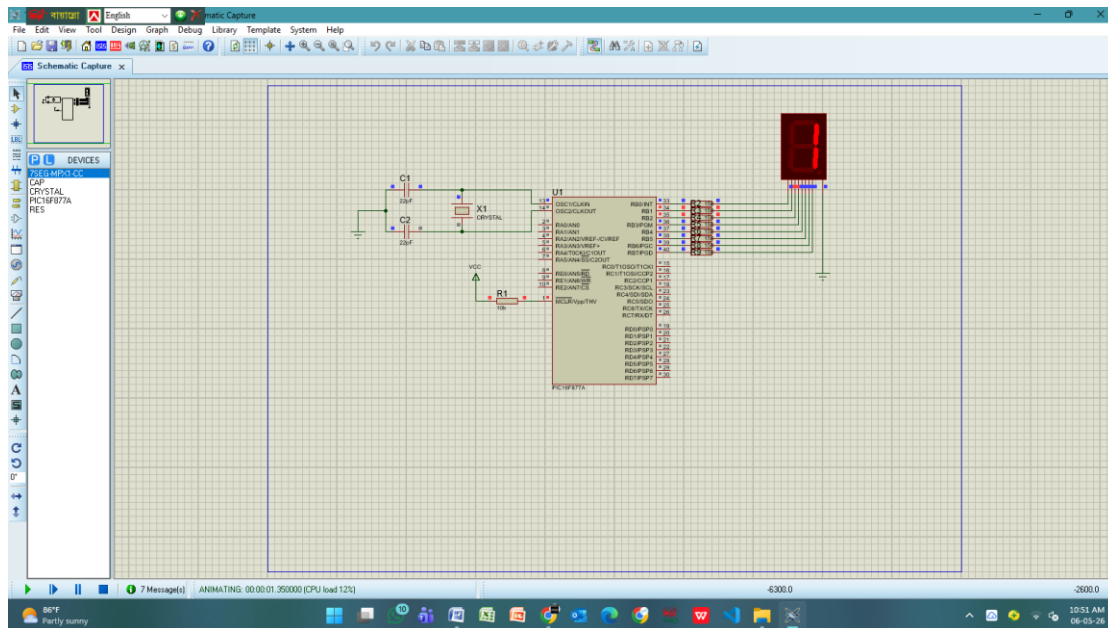
Output:



Source Code : 7 segment display

```
char arr[] = {0x3F,0x06,0x5B,0x4F,0x66,0x6D,0x7D,0x07,0x7F,0x6F};  
void main() {  
    unsigned char i;  
    TRISB = 0x00;  
    PORTB = 0x00;  
    while(1){  
        for(i = 0; i < 10; i++){  
            PORTB = arr[i];  
            Delay_ms(1000);  
        }  
    }  
}
```

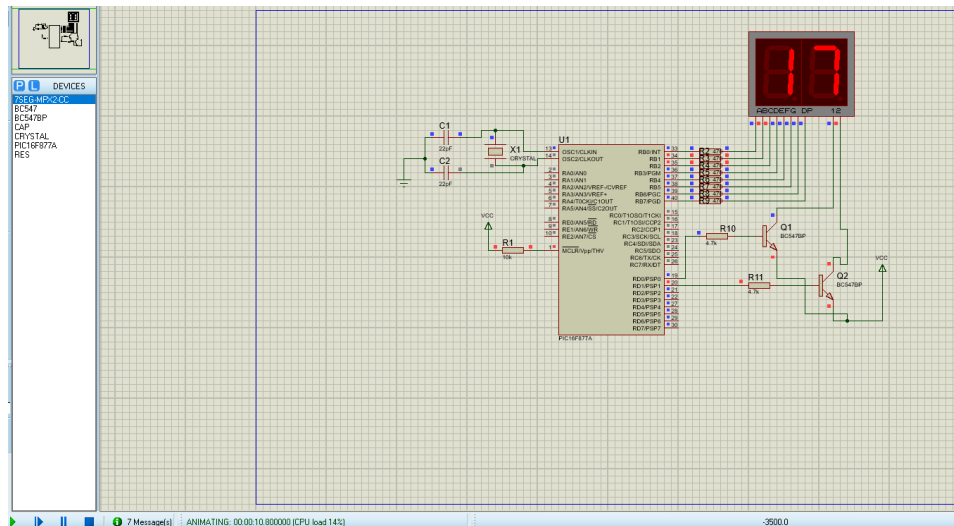
Output :



Source Code : 7 segment multiplexing

```
char arr[] = { 0x3F, 0x06, 0x5B, 0x4F, 0x66, 0x6D, 0x7D, 0x07, 0x7F, 0x6F};
int i=0;
void main() {
    int left_digit, right_digit,j;
    TRISB = TRISD = 0x00;
    portb = portd = 0x00;
    while(1){
        for(i=0; i<100;i++){    //count 99
            left_digit = i / 10;
            right_digit = i % 10;
            for(j=0;j<30; j++){
                portd.f0 = 0x00;
                portb = arr[left_digit];
                delay_ms(10);
                portd.f0=1;
                portd.f1 = 0x00;
                portb = arr[right_digit];
                delay_ms(10);
                portd.f1=1;    }    }    }
}
```

Output :



Source Code : Digital Input

```
char arr[] = {0x40,0x79,0x24,0x30,0x19,0x12,0x02,0x78,0x00,0x10};
void main() {
    int count=0,left,right;
    TRISB = TRISC = 0x00;
    TRISD = 0xff;
    PORTB = PORTC = PORTD = 0x00;
    while(1){
        if(PORTD.F0 == 1){
            Delay_ms(30);    // debounce
            if(PORTD.F0 == 1){
                count++;
                if(count == 100) count = 0;
                while(PORTD.F0 == 1); // wait till release  }
            }

            //decrease button
            if(PORTD.F1 == 1){
                Delay_ms(30);    // debounce
                if(PORTD.F1 == 1){
                    if(count == 0) count = 0;
                    else count--;
                    while(PORTD.F1 == 1); // wait till release
                }
            }

            //reset button
            if(PORTD.F2 == 1){
                delay_ms(30);
                if(PORTD.F2 == 1) {
                    count = 0;
                }
            }

            left = count/10;
            right = count%10;

            PORTC.F0 = 1;
            PORTB = arr[left]; //segment show
            Delay_ms(5);
            PORTC.F0 = 0;

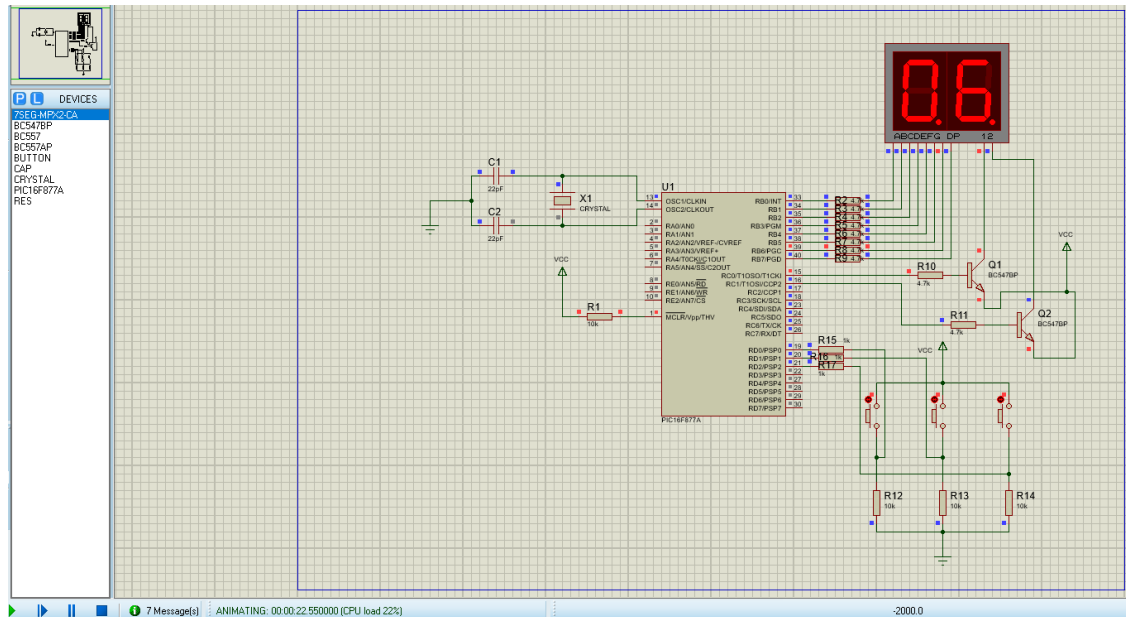
            PORTC.F1 = 1;
            PORTB = arr[right];
```

```

Delay_ms(5);
PORTC.F1 = 0;
}
}

```

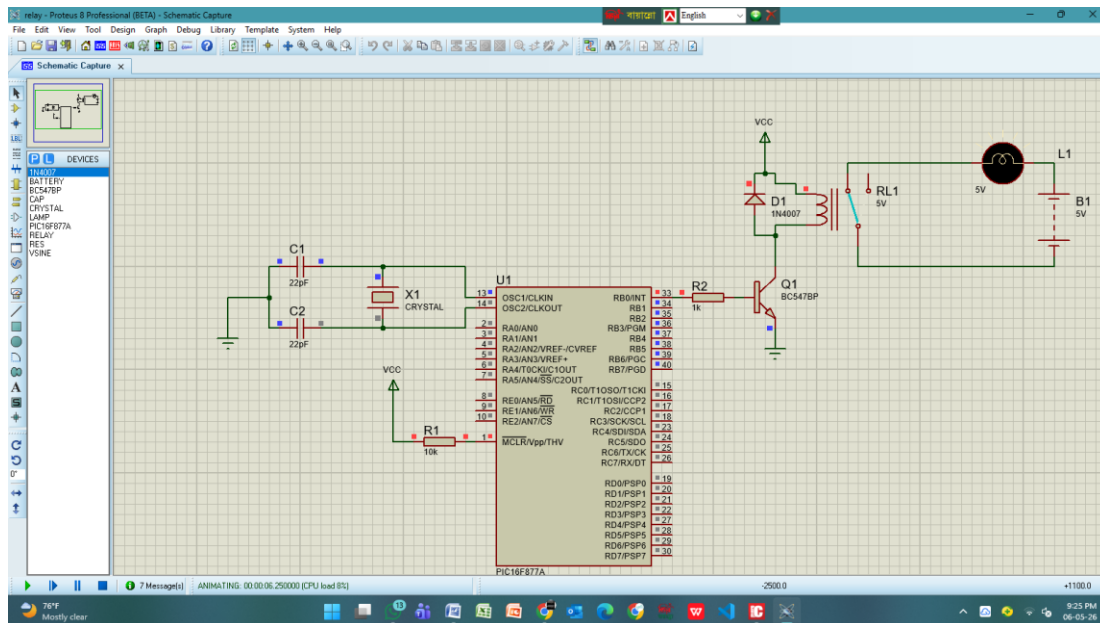
Output :



Source Code : Relay

```
void main() {  
    TRISB = 0x00; // PORTB output  
    PORTB = 0x00;  
  
    while(1) {  
        PORTB.F0 = 1; // Relay ON  
        delay_ms(1000);  
  
        PORTB.F0 = 0; // Relay OFF  
        delay_ms(1000);  
    }  
}
```

Output :



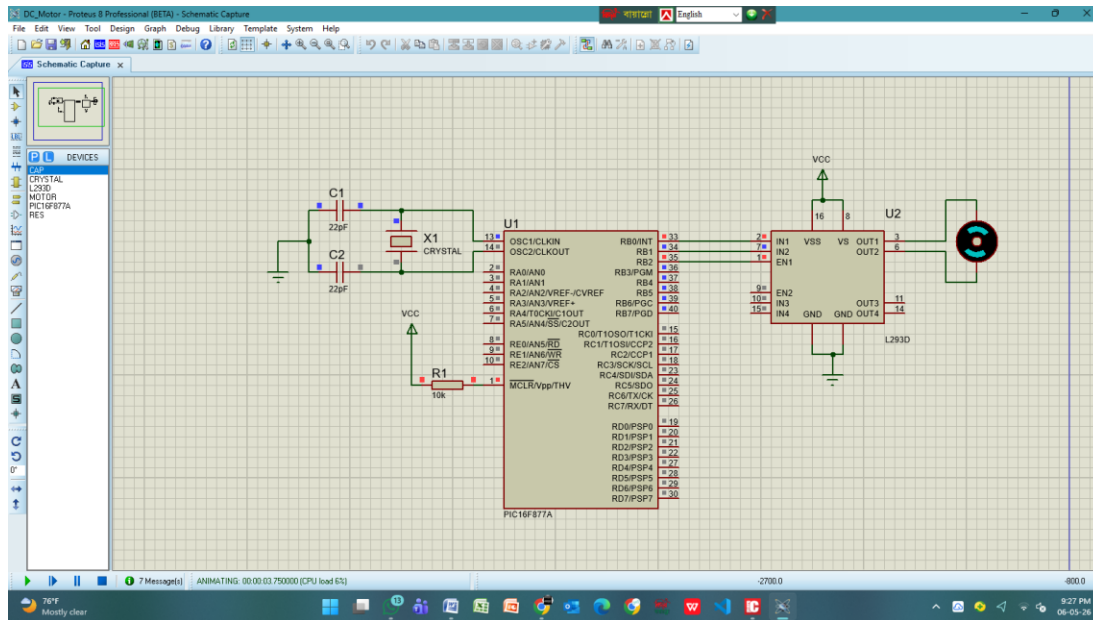
Source Code : DC motor

```
void main() {
    TRISB = 0x00;
    portb = 0x00;

    while(1){
        portb.f2=1; //en1 pin active
        portb.f0 = 1;
        portb.f1 = 0;
        delay_ms(5000);

        portb.f0 = 0;
        portb.f1 = 1;
        delay_ms(5000);
    }
}
```

Output :



Source Code : Servo Motor

```
int i;
void main() {
    TRISB.F0 = 0; // RB0 as output
    PORTB.F0 = 0;
    while(1){
        for(i=0; i<50; i++){
            PORTB.F0 = 1;
            Delay_us(1000); // 1 ms pulse
            PORTB.F0 = 0;
            Delay_us(19000); // remainder of 20 ms
        }

        Delay_ms(500);
        for(i=0; i<50; i++){
            PORTB.F0 = 1;
            Delay_us(1500); // 1.5 ms pulse
            PORTB.F0 = 0;
            Delay_us(18500);
        }

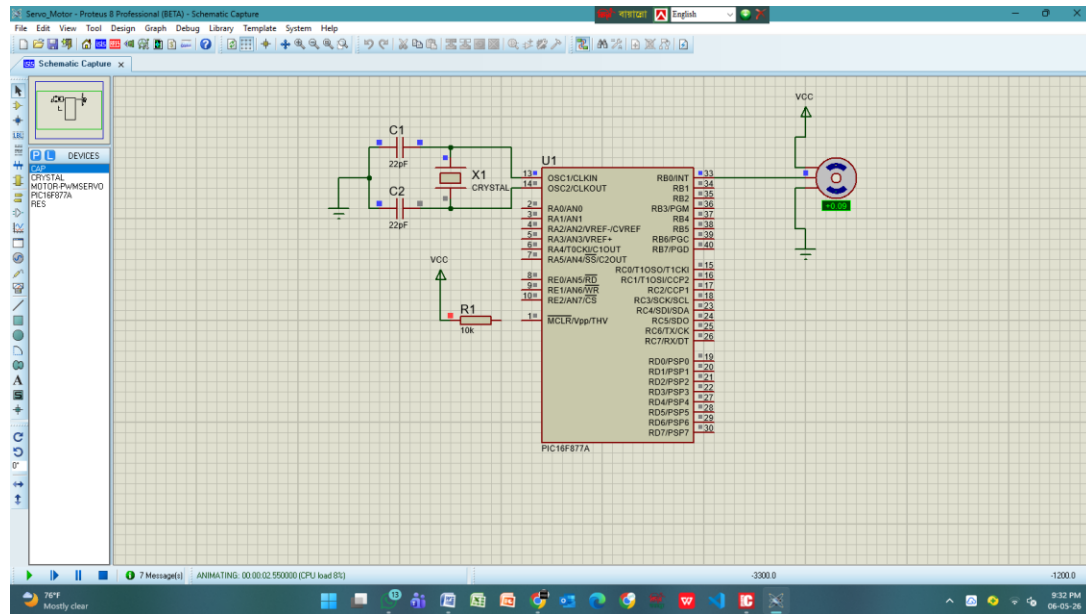
        Delay_ms(500);
        for(i=0; i<50; i++){
            PORTB.F0 = 1;
            Delay_us(2000); // 2 ms pulse
            PORTB.F0 = 0;
            Delay_us(18000);
        }

        Delay_ms(500);
        for(i=0; i<50; i++){
            PORTB.F0 = 1;
            Delay_us(1500); // 1.5 ms pulse
            PORTB.F0 = 0;
            Delay_us(18500);
        }

        Delay_ms(500);

    }
}
```


Output :



Source Code : LCD Display

```
sbit LCD_RS at RB0_bit;
sbit LCD_EN at RB1_bit;
sbit LCD_D4 at RB2_bit;
sbit LCD_D5 at RB3_bit;
sbit LCD_D6 at RB4_bit;
sbit LCD_D7 at RB5_bit;

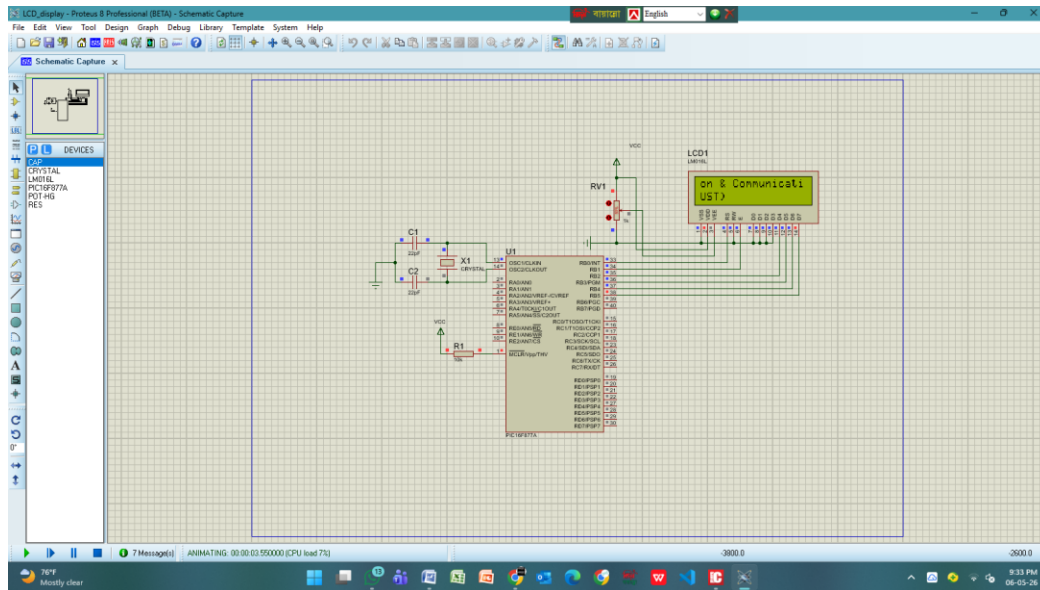
sbit LCD_RS_Direction at TRISB0_bit;
sbit LCD_EN_Direction at TRISB1_bit;
sbit LCD_D4_Direction at TRISB2_bit;
sbit LCD_D5_Direction at TRISB3_bit;
sbit LCD_D6_Direction at TRISB4_bit;
sbit LCD_D7_Direction at TRISB5_bit;
char txt1[] = "Information & Communication Engineering";
char txt2[] = "ICE(PUST)";
char i;

void Move_Delay() {
    Delay_ms(300);
}

void main(){
    ADCON1 = 0x06;
    CMCON = 0x07;

    Lcd_Init();
    Lcd_Cmd(_LCD_CLEAR);
    Lcd_Cmd(_LCD_CURSOR_OFF);
    Lcd_Out(1,1,txt1);
    Lcd_Out(2,5,txt2);
    Delay_ms(1000);
    while(1) {
        for(i=0; i<24; i++) {
            Lcd_Cmd(_LCD_SHIFT_LEFT);
            Move_Delay();
        }
    }
}
```

Output :



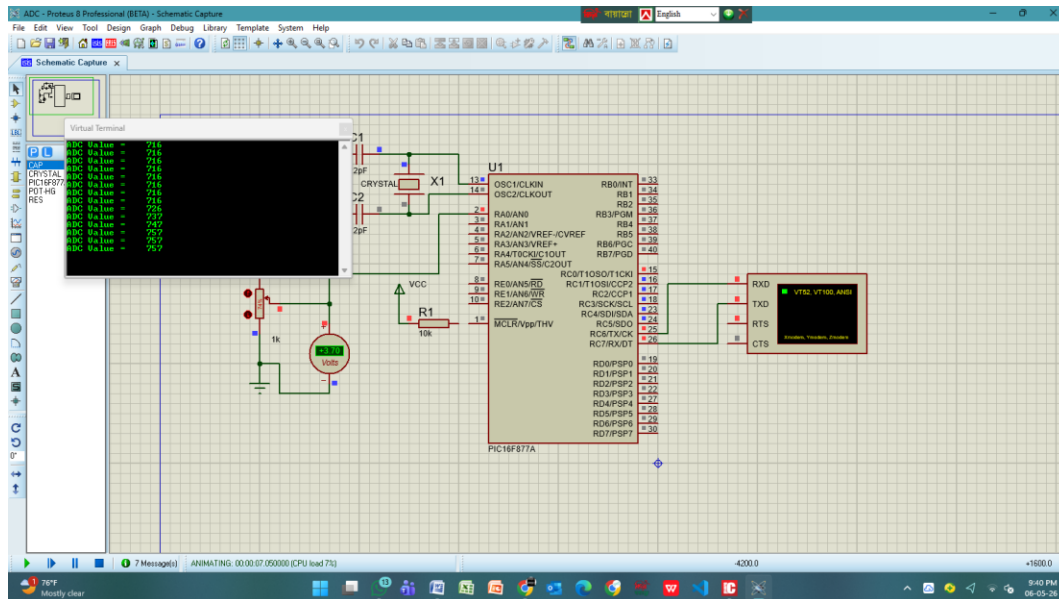
Source Code : ADC

```

unsigned int adc_value;
char txt[7];
void main() {
    ADCON1 = 0x80;      // AN0 analog
    TRISA = 0xFF;       // PORTA input (AN0 input)
    TRISC = 0x80;       // RC7 input, RC6 output
    UART1_Init(9600);    // UART 9600 baud
    Delay_ms(100);
    while(1)
    {
        adc_value = ADC_Read(0);
        IntToStr(adc_value, txt); // Integer ? String
        UART1_Write_Text("ADC Value = ");
        UART1_Write_Text(txt);
        UART1_Write(13); // Carriage return
        UART1_Write(10); // New line
        Delay_ms(500);
    }
}

```

Output :



Source Code: Temperature Sensor

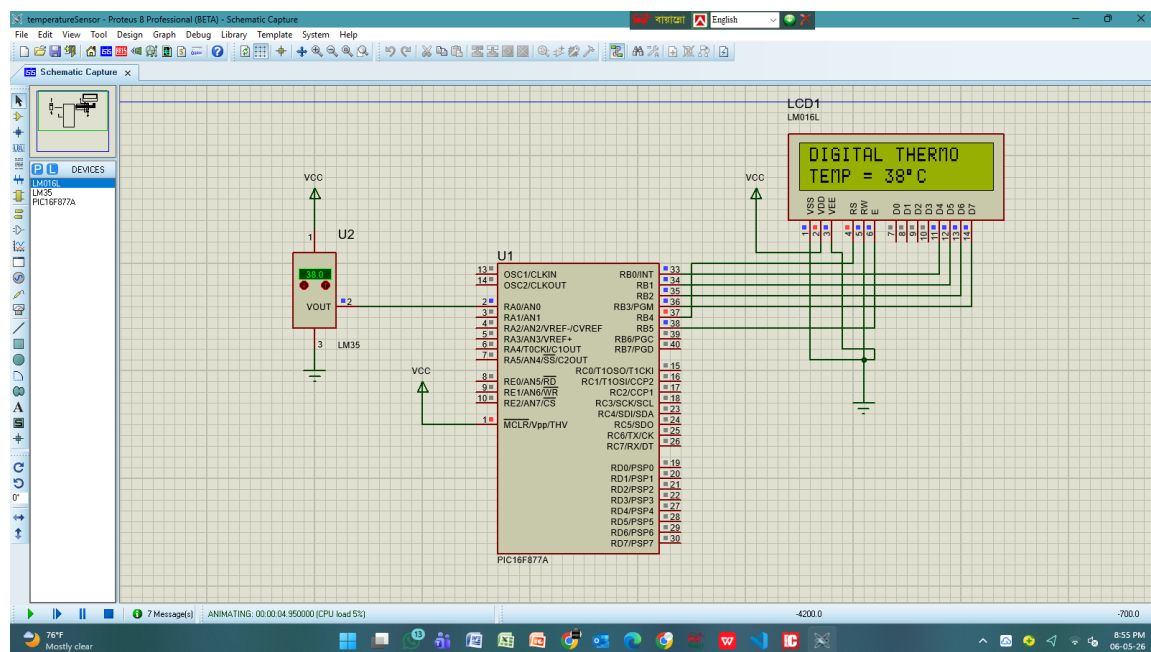
```
sbit LCD_RS at RB4_bit;
sbit LCD_EN at RB5_bit;
sbit LCD_D4 at RB0_bit;
sbit LCD_D5 at RB1_bit;
sbit LCD_D6 at RB2_bit;
sbit LCD_D7 at RB3_bit;
sbit LCD_RS_Direction at TRISB4_bit;
sbit LCD_EN_Direction at TRISB5_bit;
sbit LCD_D4_Direction at TRISB0_bit;
sbit LCD_D5_Direction at TRISB1_bit;
sbit LCD_D6_Direction at TRISB2_bit;
sbit LCD_D7_Direction at TRISB3_bit;
float temp_val;
int temp;
char temper[7];
void READ_temp(void)
{
    temp_val = ADC_Read(0);
    temp_val = temp_val * 0.48828125; // (500/1024)
    temp = (int)temp_val;
}
void data_conversion(void)
{
    IntToStr(temp, temper);
}
void display_temperature(void)
{
    Lcd_Out(2, 1, "TEMP = ");
    Lcd_Out(2, 8, Ltrim(temper));
    Lcd_Chr_Cp(0xdf); // Degree symbol
    Lcd_Chr_Cp('C');
    Lcd_Chr_Cp(' ');
}
void main()
{
    ADCON1 = 0x00;
    TRISA = 0xFF;
    ADC_Init();
    Lcd_Init();
    Lcd_Cmd(_LCD_CLEAR);
    Lcd_Cmd(_LCD_CURSOR_OFF);
    Lcd_Out(1, 1, "DIGITAL THERMO");
```

```

delay_ms(1000);
while(1)
{
    READ_temp();
    data_conversion();
    display_temperature();
    delay_ms(500);
}
}

```

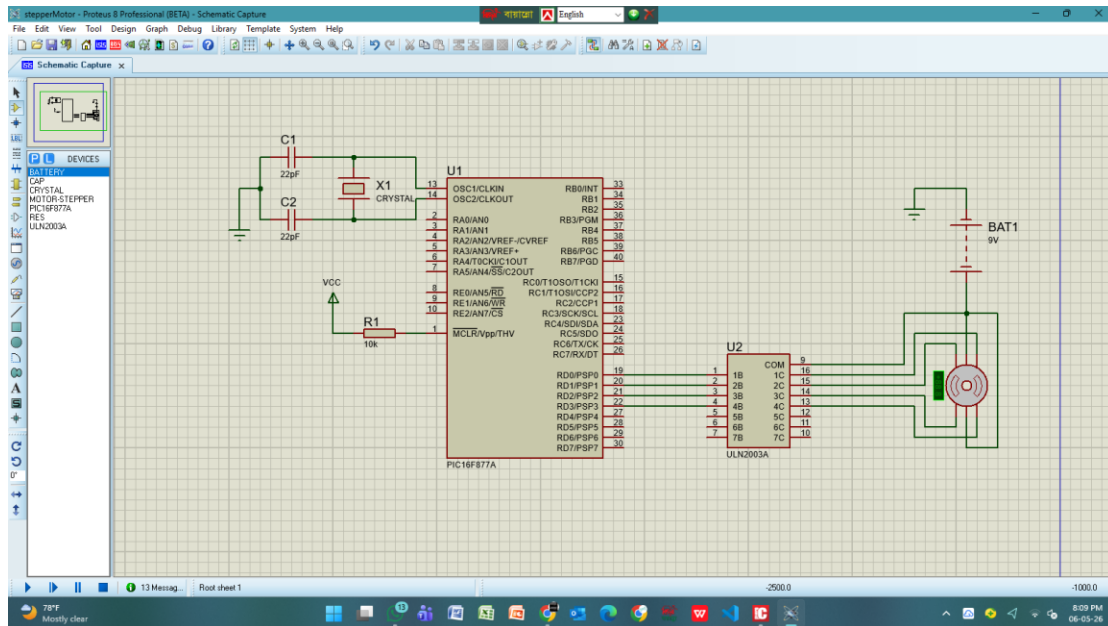
Output:



Source Code : Stepper Motor

```
void main()
{
    TRISD = 0x00;
    PORTD = 0x00;
    do
    {
        PORTD = 0b00000011;
        Delay_ms(500);
        PORTD = 0b00000110;
        Delay_ms(500);
        PORTD = 0b00001100;
        Delay_ms(500);
        PORTD = 0b00001001;
        Delay_ms(500);
    } while(1);
}
```

Output :



Source Code : EEPROM

```
unsigned char count;
unsigned char seg[] = {0x3F,0x06,0x5B,0x4F,0x66,0x6D,0x7D,0x07,0x7F,0x6F};
void main(){
    ADCON1 = 0x06;
    TRISB = 0x00;
    count = EEPROM_Read(0x00);
    if(count > 9) count = 0;
    while(1){
        PORTB = seg[count];
        Delay_ms(1000);
        count++;
        if(count > 9) count = 0;
        EEPROM_Write(0x00, count);
        Delay_ms(20);
    }
}
```

Output :

